THE MORPHISM FROM THE ZARISKI TO THE ETALE SITE IS FLAT AND CONSERVATIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof checks flatness on stalks through the strict henselization description of the etale local ring, derives exactness of epsilon^* from flatness, and proves faithfulness by tensoring at each stalk and applying faithful flatness

Editorial note. The label lemma-epsilon-flat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the localization morphism of ringed sites epsilon from the small etale site of X to its Zariski site, and the stalk formula for epsilon^*F at a geometric point
- flatness of a morphism of ringed sites checked on stalks (Modules on Sites, Lemma sites-modules-lemma-check-flat-stalks), the point morphism of sites (Sites, Lemma sites-lemma-point-morphism-sites) and enough points (Etale Cohomology, Remarks etale-cohomology-remarks-enough-points)
- the etale local ring O_{etale, xbar} is the strict henselization of O_{X,x} (Etale Cohomology, Lemma etale-cohomology-lemma-describe-etale-local-ring)
- faithful flatness of the strict henselization over the local ring (More on Algebra, Lemma more-algebra-lemma-dumb-properties-henselization)
- flat pullback of modules is exact (Modules on Sites, Lemma sites-modules-lemma-flat-pullback-exact)

Source reference. Original lemma 0GNA: sources/stacks-chow/tags.txt, source lines 20785–20785. Prerequisite pointer invoked at proof line(s) 340; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 331; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04HX: sources/stacks-chow/tags.txt, source lines 5883–5883. Prerequisite pointer invoked at proof line(s) 354; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 040S: sources/stacks-chow/tags.txt, source lines 5267–5267. Prerequisite pointer invoked at proof line(s) 352; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07QM: sources/stacks-chow/tags.txt, source lines 9837–9837. Prerequisite pointer invoked at proof line(s) 359; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05V1: sources/stacks-chow/tags.txt, source lines 7535–7535. Prerequisite pointer invoked at proof line(s) 349; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05VC: sources/stacks-chow/tags.txt, source lines 7546–7546. Prerequisite pointer invoked at proof line(s) 347; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04JC: sources/stacks-chow/tags.txt, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 364; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05V5: sources/stacks-chow/tags.txt, source lines 7539–7539. Prerequisite pointer invoked at proof line(s) 373; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The morphism ϵ of (08H7) is a flat morphism of ringed sites. In particular the functor $\epsilon^* : Mod(\mathcal{O}_X) \to Mod(\mathcal{O}_{étale})$ is exact. Moreover, if $\epsilon^*\mathcal{F} = 0$, then $\mathcal{F} = 0$.*

Proof. The flatness of the morphism ϵ is Descent, Lemma 0GNA. Here is another proof. We have to show that $\mathcal{O}_{étale}$ is a flat $\epsilon^{-1}\mathcal{O}_X$-module. To do this it suffices to check $\mathcal{O}_{X,x} \to \mathcal{O}_{étale,\overline{x}}$ is flat for any geometric point $\overline{x}$ of X, see Modules on Sites, Lemma 05VC, Sites, Lemma 05V1, and Étale Cohomology, Remarks 040S. By Étale Cohomology, Lemma 04HX we see that $\mathcal{O}_{étale,\overline{x}}$ is the strict henselization of $\mathcal{O}_{X,x}$. Thus $\mathcal{O}_{X,x} \to \mathcal{O}_{étale,\overline{x}}$ is faithfully flat by More on Algebra, Lemma 07QM.

The exactness of ϵ^* follows from the flatness of ϵ by Modules on Sites, Lemma 04JC.

Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. If $\epsilon^*\mathcal{F} = 0$, then with notation as above

$$0 = \epsilon^*\mathcal{F}_{\overline{x}} = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{étale,\overline{x}}$$

(Modules on Sites, Lemma 05V5) for all geometric points $\overline{x}$. By faithful flatness of $\mathcal{O}_{X,x} \to \mathcal{O}_{étale,\overline{x}}$ we conclude $\mathcal{F}_x = 0$ for all $x \in X$. □